

greater than for a house; as such, collateralization ratios for the ETH–DAI contract are higher than a traditional mortgage. In addition, no centralized institution is necessary because everything happens within the Ethereum blockchain.

Let's consider a simple example. Suppose an ETH owner needs liquidity but does not want to sell her ETH because she thinks it will appreciate. The situation is analogous to the homeowner who needs liquidity but does not want to sell their house. Let's say an investor has 5 ETH at a market price of \$200 (total value of \$1,000). If the collateralization requirement is 150 percent, then the investor can mint up to 667 DAI ($\$1,000/1.5$ with rounding). The collateralization ratio is set high to reduce the probability that the loan debt exceeds the collateral value. In addition, for the DAI token to be credibly pegged to the USD, the system needs to avoid the risk that the collateral is worth less than $\$1 = 1$ DAI.

Given the collateralization ratio of 1.5, it would be unwise to mint the 667 DAI because if the ETH ever dropped below \$200 the contract would be undercollateralized – the equivalent of a *margin call*. We are using traditional finance parlance, but in DeFi there is no communication from your broker about the need to post additional margin or to liquidate the position and also no grace period. Liquidation can happen immediately.

As such, most investors choose to mint less than 667 DAI to give themselves a buffer. Suppose the investor mints 500 DAI, which implies a collateralization ratio of 2.0 ($\$1,000/2.0 = 500$). Let's explore two scenarios. First, suppose the price of ETH rises by 50 percent so that the